Differential Gene Expression in the Aging Brain

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1 Introduction

In this document, we detail our analyses of the aging brain microarray data. We begin with exploratory data analysis and identify several sources of systematic bias in the dataset. We describe our procedure for normalizing the data. We identify differentially expressed genes using linear modeling and classification methods and conclude that there exist technical imperfections in the data et that need to be corrected. Finally, as a subsequent analysis, we change our age cutoff and supply a list of candidate genes for future confirmatory assays.

2 Exploratory Data Analysis

The exploratory analysis aimed to identify structural biases in the dataset that could confound the discovery of differentially expressed (DE) genes across age and brain regions. Several imbalances were detected across demographic and technical variables.

Age distribution bias. The number of samples with age < 70 was substantially larger than those with age ≥ 70, as shown in Figure 1. This imbalance was preserved across brain regions, implying that the younger group contributes disproportionately to variability in the data. Consequently, DE detection in the older group may suffer from reduced power.

age_group	<70	>=70
region		
ancg	104	23
cb	70	11
dlpfc	146	32

(a) Counts by age group within regions

lab	davis		irvine		michigan	
arrayversion	1.0	2.0	1.0	2.0	1.0	2.0
region						
ancg	0	0	36	30	31	30
cb	11	10	33	9	9	9
dlpfc	56	29	9	0	55	29

(b) Counts by region within (lab,version)

Figure 1: Sample-count distributions that reveal strong design imbalance.

Brain-region representation and lab imbalance. Figure 1 shows that each brain region is associated predominantly with specific laboratory sites and array versions. For instance, ANCg samples were processed only at Irvine and Michigan labs, while DLPFC samples were absent from Irvine v2. Such concentration of regions within specific labs introduces potential *batch effects* that may be confounded with biological differences.

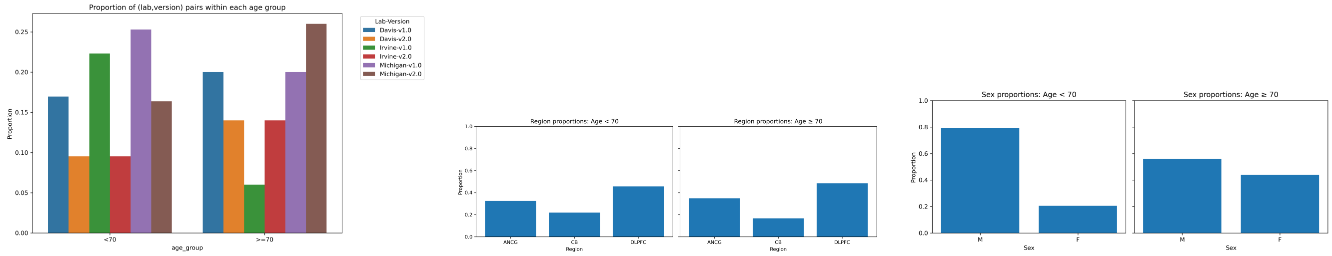
Effect of lab and array version on age groups. Figure 2 demonstrates that certain (lab, arrayversion) combinations dominate specific age groups. Michigan v2 and Davis v1 are more prevalent in older subjects, whereas Irvine v1 and Michigan v1 are overrepresented in younger subjects. This technical imbalance could mimic or mask age-related expression trends.

Sex imbalance. Figure 2 shows that the younger age group is overwhelmingly male, whereas the older group is more balanced. This imbalance increases the risk of detecting sex-driven rather than age-driven expression differences. Later cosine similarity analyses (in the Appendix) confirm a moderate association between sex and expression profiles.

Region and technical effects on expression. Standardized cosine similarity analyses (described in the appendix) reveal strong clustering by brain region and consistent patterns by lab and array version. In particular, the Davis v2 samples show a distinct technical signature even among control probes, confirming a measurable lab–version effect.

Recommendations to reduce bias in future studies

- Apply balanced sampling during experiment design (equal representation of labs, versions, sexes, regions and age groups).
- Use randomization to avoid coupling of region or age with technical covariates.



(a) (lab,version) within age

(b) Region within age

(c) Sex within age

Figure 2: Composition within age groups: technical and biological covariates differ across age.

3 Linear Modeling, Normalization, and Results

Applying the linear modeling techniques detailed in the appendix (Section 8.2), we began looking for differentially expressed genes. Part of the value of these modeling techniques is the identification of bias in the dataset. An example includes the fitting of a linear model involving region, sex, lab, age group, array version, and the interaction between lab and array version. When we plot contrasts between labs for array version 2 and contrasts in brain region (the latter involving a model which excludes the interaction term), holding the other variables constant, we clearly see that genes tend to be expressed more in array version 2 of some labs and in some brain regions. See figure 4 for an illustration. This obscures the differential expression of genes in some brain regions than in others. We apply our normalization procedure to correct for technical biases in the dataset. See figure 5 to see the results of this correction.

After normalizing the data, we sought to identify differentially expressed genes. See figure 6 for the corresponding MA plots and volcano plots for the entire age comparison and the age comparison within the dorsolateral prefontal cortex. After normalizing and fitting the contrasts between age groups in each region, we produced a list of candidate differentially expressed genes. See figures 7 and 8 for an abridged list of these genes.

We subsequently performed PCA to identify unusual individuals in the dataset. In particular, we identified the top 50 principal components for the gene expression data. We found that the first two principal components accounted for roughly 92% of the variation in the dataset. See figure 9. In this plot we identified a highly unusual individual in the principal component space. We identified the individual as a 65-year-old female. Given the unusual nature of her gene expression, we decided to analyze the robustness of our results by changing the age cutoff from 70 to 65. The results of this change are described in sections 6 and 7, respectively.

4 Age-Predictive Linear Modelling

The rationale behind this modeling framework was that **genes and covariates which are predictive of age are likely to be biologically relevant candidates for differential expression (DE)** between younger and older individuals. To formalize this, we employed a **supervised linear modelling pipeline** designed to identify covariates (genes and interaction terms) most strongly associated with chronological age. The workflow included a standard train-test split, followed by variable selection on the train set using correlation with age and LASSO. The variables considered included genes as well as arraymeta variables and interaction between the two. Then we ran OLS regression on the test split. After that the (Benjamini-Hochberg corrected) significant covariates were tested by a 2 sample T-test within each brain region to identify DE by age.

The resulting BH-corrected significant features are shown below. The context column shows if the gene was identified as an interaction term with some arraymeta variable. If yes, the 2-sample t-tests were run within each region as well as on the subsets of the data within each region that corresponded to the interacting arraymeta variable. If the context says ”arraymeta variable = value”, it means that the interaction term of arraymeta variable with the gene was found to be significant and the gene was found to be significant in the t-test *without* taking subset within the region. If the context says ”arraymeta variable = value | subset:arraymeta variable=value”, it means that the gene was found to be significant in the t-test but by taking subset within the region. This can indicate a potential confounding effect due to the arraymeta variable.

DLPFC (dorsolateral prefrontal cortex) emerged as the only region with significant DE genes under this framework. This outcome is biologically coherent: DLPFC is strongly implicated in **cognitive and memory processes**, which exhibit well-documented age-related decline. Thus, the linear model effectively captured the predominant pattern of age-associated transcriptional change.

Differential Expression by Region (BH significant)

Gene	Probe	Region	Context	n<70	n≥70	t	p	q
ERBB2IP	217941_s_at	DLPFC	arrayv=1 | subset: arrayversion=1	98	22	-5.148	0.0	0.0
FAM107A	209074_s_at	DLPFC	arrayv=1 | subset: arrayversion=1	98	22	-3.517	0.001	0.044
nan	208451_s_at	DLPFC		146	32	-3.319	0.002	0.048
IFI16	206332_s_at	DLPFC	sex=M	146	32	-3.257	0.002	0.048
PLSCR4	218901_at	DLPFC	sex=M	146	32	-3.162	0.003	0.052
PTPRO	211600_at	DLPFC		146	32	-2.994	0.004	0.058
FTL	212788_x_at	DLPFC	region.dlpfc	146	32	-2.961	0.004	0.058
NFE2L2	201146_at	DLPFC	sex=M	146	32	-2.869	0.006	0.07
TNS1	221748_s_at	DLPFC		146	32	-2.757	0.008	0.075
CHI3L1	209395_at	DLPFC		146	32	-2.758	0.008	0.075
QKI	212636_at	DLPFC	sex=M | subset: sex=M	115	17	-2.787	0.009	0.075
QKI	212636_at	DLPFC	sex=M	146	32	-2.703	0.009	0.075
PLSCR4	218901_at	DLPFC	sex=M | subset: sex=M	115	17	-2.786	0.011	0.08
ERBB2IP	217941_s_at	DLPFC	arrayv=1	146	32	-2.591	0.012	0.083
SLC7A11	209921_at	DLPFC	sex=M | subset: sex=M	115	17	-2.706	0.013	0.083
NFE2L2	201146_at	DLPFC	sex=M | subset: sex=M	115	17	-2.601	0.015	0.09
LGALS3	208949_s_at	DLPFC		146	32	-2.474	0.016	0.093

Figure 3: Significant genes and covariates identified from the age-predictive linear modelling pipeline

5 Classification

Along the age predictive regression modeling, we also consider the classification modeling of age < 70 and ≥ 70. We fit a ridge-regularized logistic model using all non-control genes as predictors. Gene expressions were column-wise standardized, and we included region/lab/version one-hots as covariates to take batch effects into account. Hyperparameters were tuned by cross-validation, and performance was summarized with ROC-AUC, balanced accuracy, and the confusion matrix. See Appendix for more detailed description.

We ranked genes by the absolute coefficient in Figure 20. Results show the top 20 influential genes in this classifier. To comment on the reliability of this result, we should notice that the "importance" defined here depends on the model and is not robust to the choice of classification algorithms. Moreover, the model we considered does not condition on regions.

To address this concern, changing the age cutoff could provide some robustness. If a gene is confirmed to be important again by a different classification model, we would have more confidence in the significance of the gene. Instead of other classification algorithms, we consider the same approach but with different age cutoffs and expect some overlap in the identified gene sets.

6 Changing Age Cutoff

To determine how robust our analysis was, we changed the age cutoff from 70 to 65. The results of this change in cutoff are illustrated in figures 10, 11, 12, 13. Using the limma technique, we found that **GRIK1, IFT122**, and **SPR** were found in both gene lists when considering all of the twenty most significant genes. When considering the cerebellum, we found that genes **HSD11B1, PENK, RPS6KB1, XIST** were found in both gene lists, with the presence of **XIST** suggesting that there is some confounding with gender. Each figure lists the genes present on both lists.

We did this for the age predictve linear model as well. Using BH at 0.1, we compared DE genes with age cutoffs of 70 vs 65. Of the **13** unique genes at the 70 cutoff, **11** are also significant at 65 (**84.6%** retention), indicating broadly robust findings to the age threshold. The shared set includes **ERBB2IP, FAM107A, IFI16, PLSCR4, PTPRO, FTL, NFE2L2, CHI3L1, QKI, SLC7A11,** and probe `208451_s_at` (no symbol listed). The list of genes for this cutoff can be found in the Appendix, Figure 17.

Two genes exhibit significance both overall and within a subset defined by an array-metadata indicator, suggesting that apparent effects can be partly driven by technical or demographic composition: **ERBB2IP** (with arrayversion=1) and **PLSCR4** (with sex=M) were common to both cutoffs. These patterns underscore that array version and sex can modulate detectability and should be adjusted when interpreting region-specific DE results. As a result, we do not include them in our final recommendation for further experimentation.

The result given by the ridge logistic approach is given in Figure 21. The shared genes in both cutoff 65 and 70 are **S100A8, WIF1, TMEM144, GFAP** and **GPR17**.

7 DE Genes for Future Analyses

Our final recommendation of genes that should be tested for differential expression is the compilation of results from all 3 of our models. We recommend **19** genes to be tested.

For testing exclusively in DLPFC, we suggest **FAM107A, IFI16, PTPRO, FTL, NFE2L2, CHI3L1, QKI, SLC7A11**. The unnamed probe `208451_s_at` is also significant in DLPFC but should be investigated for gene identification first. For the cerebellum, we recommend testing the genes **HSD11B1, PENK** and **RPS6KB1**.

Along with this, some genes showed significant overall differential expression but not in any particular region. So we recommend that **S100A8, WIF1, TMEM144, GFAP, GPR17, GRIK1, IFT122** and **SPR** be tested in all regions to gather more data that will help in removing potential confounding effects between region and other array metadata.

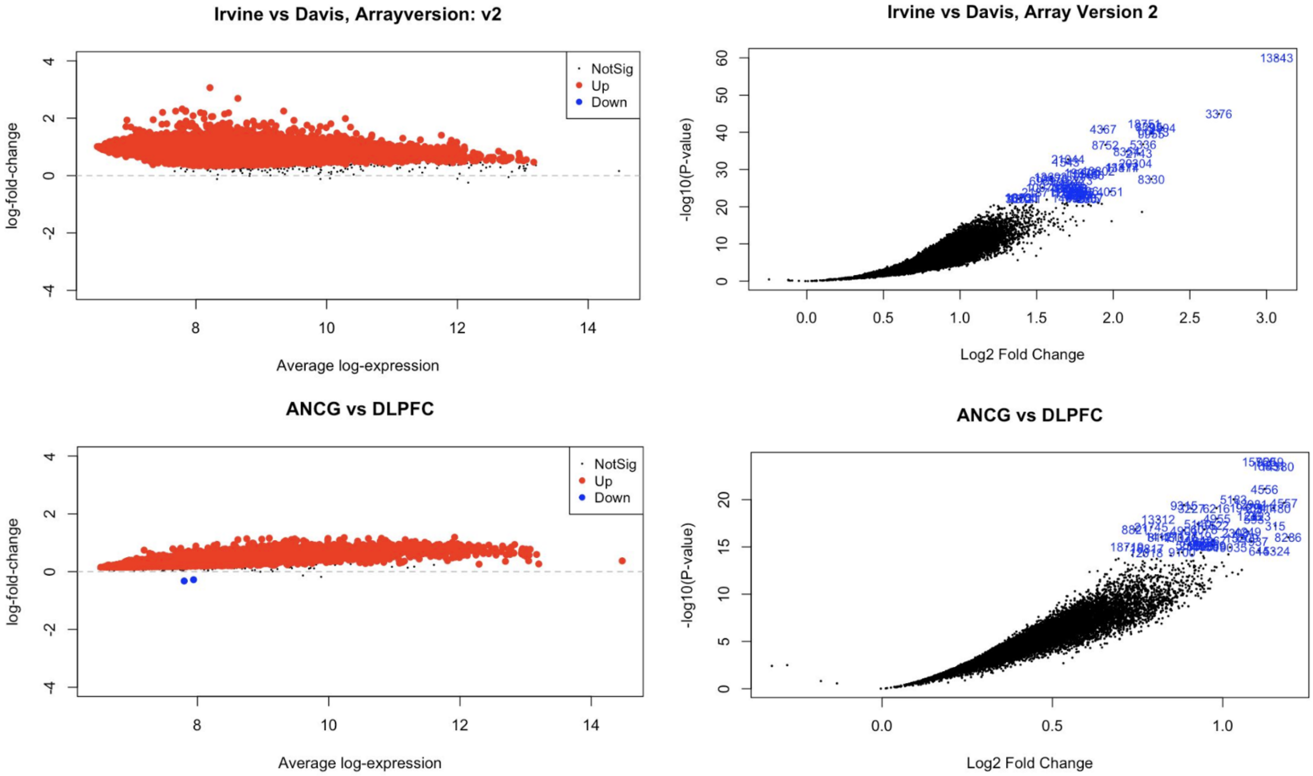


Figure 4: Systematic Bias in Gene Expression in Labs and Array Versions

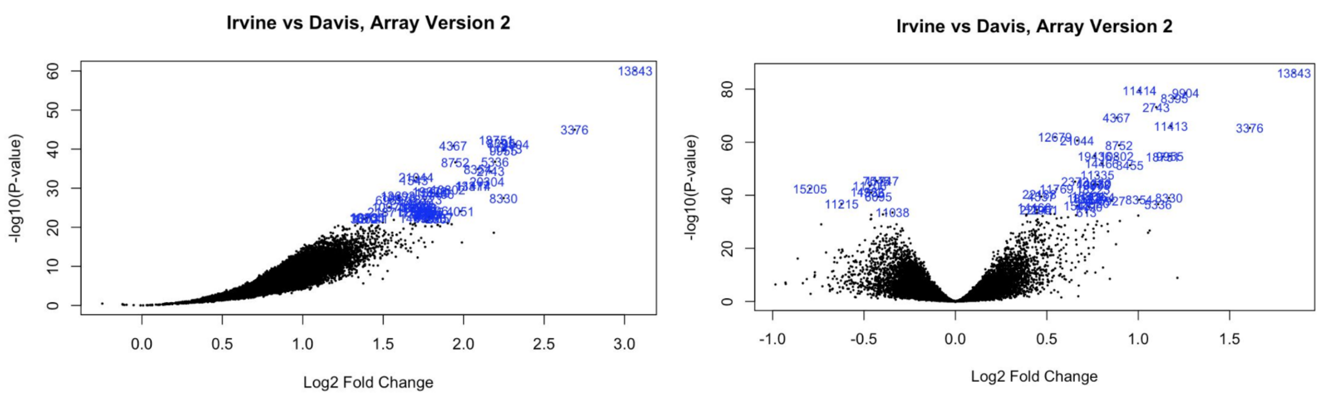


Figure 5: Correction of technical bias using our normalization procedure

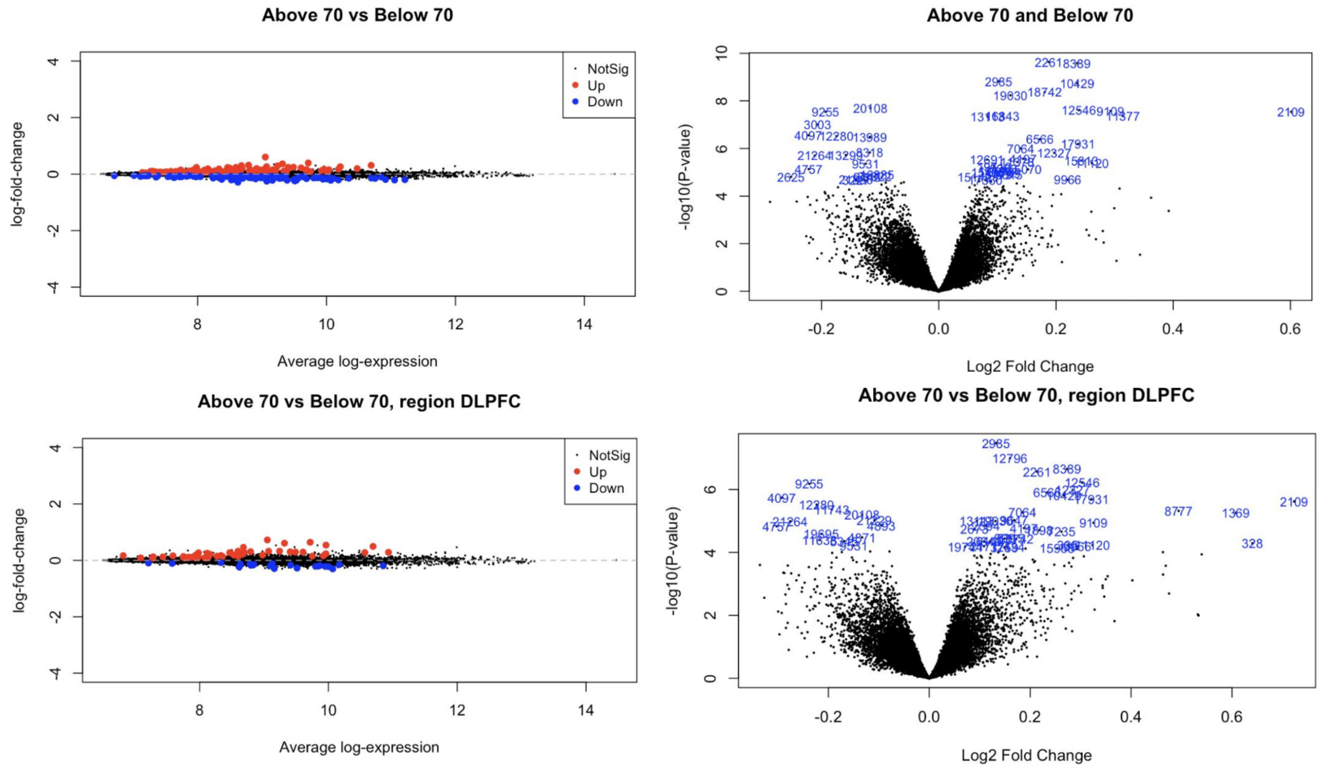


Figure 6: MA and Volcano plots for contrasts of interest

P4HA2 : Linked to poor prognosis in several cancers, including breast, oral, lung, and thyroid cancers. Promotes tumor cell invasion, migration, and metastasis

HLA-DRA : Altered HLA-DRA associated with autoimmune disease and immune escape in cancers (such as ovarian/cervical cancers)

SPR : Involved in neurotransmitter production and movement disorders like dystonia/involuntary movements (cerebellar diseases)

NRG1 : A family of proteins involved in cell-cell signaling; also implicated in certain types of cancer, such as non-small cell lung cancer

MS4A6A : Implicated in various cell functions, particularly in microglia (cells involved in CNS to remove plaques, damaged/unnecessary neurons, etc)

Figure 7: Some differentially expressed genes across all regions. Red indicates upregulation in the above 70 group as opposed to the less than 70 group.

Anterior Cingulate Cortex

TPBG : encodes the 5T4 oncofetal antigen - glycoprotein involved in cell adhesion, migration, and development. Influences signaling pathways in cancer

CD74 : contributes to pro-inflammatory tumor microenvironment - acts as a survival receptor in various cancers, notably clear cell renal cell carcinoma and non-small cell lung cancer

Dorsolateral Prefrontal Cortex

SPR : (From above)

CLIC2 : encodes diverse group of proteins that regulate fundamental cellular processes including stabilization of cell membrane potential

HLA-DRA : (From above)

P4HA2 : (From above)

Cerebellum

PENK : Modulates production of pentapeptide opioids released into the synapse where they bind to mu- and delta-opioid receptors to modulate the perception of pain

CACNA1G : provides instructions for making a low-voltage-activated calcium channel, which play crucial roles in heart pacemaker activity, brain function, and cell growth

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Figure 8: Some differentially expressed genes in each region. As before, red indicates upregulation in above 70 as opposed to less than 70 group. Meanwhile, blue indicates downregulation in above 70 as opposed to the less than 70 group.

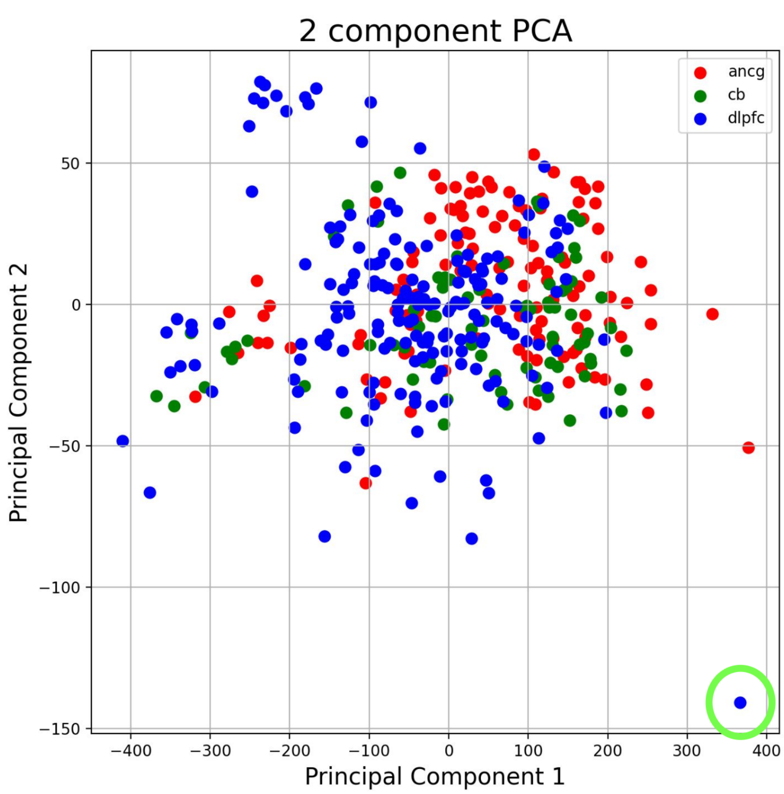


Figure 9: Two-component PCA and the identification of an unusual individual in the dataset (green circle). The points of the principal component are colored according to the brain region. For example, the individual in question had her dorsolateral prefontal cortex sampled.

	logFC	AveExpr	t	P.Value	adj.P.Val	B	gene
0	0.187647	8.158321	6.516286	2.326005e-10	0.000003	13.084611	P4HA2
1	0.235723	8.464806	6.497163	2.607713e-10	0.000003	12.977845	HLA-DRA
2	0.102138	8.353890	6.197900	1.509143e-09	0.000010	11.339609	SPR
3	0.236140	7.981742	6.166851	1.804105e-09	0.000010	11.173184	HLA-DRA
4	0.180672	8.331194	6.014910	4.278881e-09	0.000019	10.368544	NARG1L
5	0.122201	8.123996	5.959722	5.831321e-09	0.000022	10.080332	MS4A6A
6	-0.116556	8.632771	-5.733369	2.027753e-08	0.000059	8.921147	IFT122
7	0.239060	9.406320	5.696690	2.472704e-08	0.000059	8.736811	SLC5A3
8	0.293073	9.240767	5.683650	2.652760e-08	0.000059	8.671513	CD74
9	0.601215	9.050568	5.668326	2.880719e-08	0.000059	8.594936	NaN
10	-0.193073	8.613387	-5.667648	2.891223e-08	0.000059	8.591556	NaN
11	0.108521	7.971973	5.595974	4.241376e-08	0.000073	8.235728	HLA-DMA
12	0.314095	8.790392	5.591381	4.346250e-08	0.000073	8.213055	HLA-DPA1
13	0.083269	7.932206	5.582059	4.566897e-08	0.000073	8.167090	MYO1F
14	-0.206698	8.523121	-5.435604	9.858067e-08	0.000146	7.453382	TPBG
15	-0.222349	9.158630	-5.232180	2.793643e-07	0.000389	6.488823	COX7A1
16	-0.174525	10.003463	-5.216589	3.021858e-07	0.000396	6.416191	ABR
17	-0.117626	7.644869	-5.199710	3.289275e-07	0.000407	6.337773	GRIK1
18	0.172673	9.960131	5.156717	4.078197e-07	0.000478	6.139020	ST13
19	0.237675	8.593706	5.064332	6.441565e-07	0.000717	5.716745	CHORDC1

	logFC	AveExpr	t	P.Value	adj.P.Val	B	gene
0	-0.115599	7.644869	-6.132270	2.199106e-09	0.000049	11.046550	GRIK1
1	0.172445	7.980030	5.848053	1.083479e-08	0.000121	9.543596	NaN
2	-0.260798	8.833856	-5.610306	3.929611e-08	0.000292	8.331802	NPTX2
3	-0.125701	7.977471	-5.526926	6.112752e-08	0.000326	7.916799	C18orf10
4	0.422200	11.992566	5.492418	7.327980e-08	0.000326	7.746581	GFAP
5	0.100478	8.359324	5.337062	1.639248e-07	0.000609	6.991460	PHF20
6	-0.192628	9.512150	-5.236087	2.739078e-07	0.000780	6.510591	FAM131A
7	0.073775	8.353890	5.231643	2.801174e-07	0.000780	6.489606	SPR
8	-0.142399	9.324676	-4.957937	1.081484e-06	0.002409	5.227024	GGCT
9	-0.111451	9.512708	-4.957928	1.081532e-06	0.002409	5.226982	PTP4A1
10	0.172286	8.517328	4.908281	1.373168e-06	0.002501	5.004294	LOC399491
11	-0.094632	7.719527	-4.887073	1.519698e-06	0.002501	4.909765	HRASLS
12	-0.071644	7.846070	-4.885227	1.533148e-06	0.002501	4.901551	TNNI3K
13	-0.221455	10.185070	-4.880018	1.571706e-06	0.002501	4.878398	PCP4
14	-0.195639	9.492886	-4.841771	1.884953e-06	0.002648	4.709041	HOMER1
15	0.102251	7.926211	4.839856	1.902128e-06	0.002648	4.700591	POMZP3
16	0.058684	7.761296	4.776455	2.564029e-06	0.003360	4.422536	SPAG5
17	-0.141991	8.515203	-4.735330	3.106691e-06	0.003722	4.243904	WTAP
18	-0.082253	8.632771	-4.730695	3.174385e-06	0.003722	4.223854	IFT122
19	-0.212806	8.136616	-4.719123	3.349621e-06	0.003731	4.173879	CRH

Figure 10: Contrasting gene expression results across all regions for different age cutoffs. Genes in both lists: GRIK1, IFT122, SPR

	logFC	AveExpr	t	P.Value	adj.P.Val	B	gene
0	-0.318639	8.523121	-5.202556	3.259876e-07	0.007262	5.901806	TPBG
1	0.172403	7.687387	4.694981	3.759375e-06	0.041874	3.817409	COL3A1
2	0.258352	8.464806	4.288037	2.301694e-05	0.153998	2.281870	HLA-DRA
3	0.142447	8.123996	4.174883	3.719273e-05	0.153998	1.876958	MS4A6A
4	-0.226933	8.354725	-4.172102	3.762903e-05	0.153998	1.867129	CUX2
5	0.137379	8.216330	4.135300	4.388171e-05	0.153998	1.737614	HRH1
6	0.377880	8.790392	4.088731	5.321907e-05	0.153998	1.575209	HLA-DPA1
7	0.347666	9.240767	4.079407	5.530300e-05	0.153998	1.542892	CD74
8	0.167699	8.684834	4.006397	7.452268e-05	0.184460	1.292153	SCPEP1
9	0.139491	7.330984	3.797912	1.704527e-04	0.379717	0.598855	PLN
10	0.170903	7.968581	3.688844	2.589922e-04	0.477348	0.249687	HLA-DPA1
11	0.113069	7.589987	3.651276	2.984439e-04	0.477348	0.131584	ADAM28
12	0.105209	7.478111	3.639743	3.116458e-04	0.477348	0.095553	C7
13	-0.197344	8.240946	-3.637824	3.138953e-04	0.477348	0.089568	DIO2
14	0.231341	7.981742	3.631499	3.214178e-04	0.477348	0.069861	HLA-DRA
15	0.152870	8.966916	3.529329	4.690026e-04	0.564349	-0.244072	IGHG1
16	-0.109021	9.219220	-3.526138	4.745030e-04	0.564349	-0.253743	IK
17	-0.132206	8.949969	-3.525126	4.762596e-04	0.564349	-0.256807	CTSL1
18	0.107388	7.562958	3.522224	4.813317e-04	0.564349	-0.265593	TAS2R16
19	0.203036	8.597845	3.492270	5.367241e-04	0.574023	-0.355886	APLNR

	logFC	AveExpr	t	P.Value	adj.P.Val	B	gene
0	-0.148011	7.977471	-4.109290	0.000049	0.771928	1.737946	C18orf10
1	0.139478	7.926211	4.024273	0.000069	0.771928	1.434726	POMZP3
2	0.123428	7.824409	3.718796	0.000231	0.995427	0.393072	NaN
3	0.130425	8.966916	3.502226	0.000518	0.995427	-0.299455	IGHG1
4	-0.076231	7.963389	-3.496968	0.000528	0.995427	-0.315790	GEMIN6
5	-0.100062	8.434893	-3.478077	0.000565	0.995427	-0.374285	BID
6	-0.152543	8.540425	-3.471984	0.000578	0.995427	-0.393089	LOC100129361
7	0.067493	7.797037	3.415380	0.000707	0.995427	-0.566312	NaN
8	-0.156855	8.354725	-3.347484	0.000899	0.995427	-0.770577	CUX2
9	0.091539	7.409974	3.340898	0.000920	0.995427	-0.790187	PPBP
10	0.119349	8.684834	3.315434	0.001005	0.995427	-0.865667	SCPEP1
11	0.098767	8.745970	3.313436	0.001012	0.995427	-0.871565	NEK9
12	0.066663	7.995249	3.303992	0.001046	0.995427	-0.899407	KIR2DL3
13	0.104353	7.687387	3.291757	0.001091	0.995427	-0.935363	COL3A1
14	0.174670	7.896728	3.280837	0.001133	0.995427	-0.967349	IFI44L
15	0.079420	8.549476	3.262502	0.001207	0.995427	-1.020833	SMYD5
16	0.064993	7.511007	3.248835	0.001265	0.995427	-1.060514	HHAT
17	-0.173427	8.523121	-3.246824	0.001273	0.995427	-1.066340	TPBG
18	0.076835	7.428941	3.245700	0.001278	0.995427	-1.069596	ADH1B
19	0.123682	8.885776	3.236650	0.001318	0.995427	-1.095762	BAIAP3

Figure 11: Contrasting gene expression results for brain region AnCG for different age cutoffs. Genes in both lists: COL3A1, CUX2, IGHG1, SCPEP1, TPBG

	logFC	AveExpr	t	P.Value	adj.P.Val	B	gene
0	0.132402	8.353890	5.641225	3.355312e-08	0.000747	8.266490	SPR
1	0.160038	7.585101	5.431780	1.011864e-07	0.001127	7.273772	CLIC2
2	0.273049	8.464806	5.278380	2.223675e-07	0.001510	6.566767	HLA-DRA
3	0.213737	8.158321	5.239158	2.711741e-07	0.001510	6.388752	P4HA2
4	0.303053	9.406320	5.078182	6.046738e-07	0.002448	5.670052	SLC5A3
5	-0.237782	8.613387	-5.060562	6.593427e-07	0.002448	5.592554	NaN
6	0.284244	8.690098	4.970068	1.024498e-06	0.003260	5.198196	SLC5A3
7	0.233967	9.960131	4.930809	1.237875e-06	0.003447	5.029030	ST13
8	0.267179	7.981742	4.884835	1.542510e-06	0.003818	4.832408	HLA-DRA
9	-0.292302	9.158630	-4.849172	1.827474e-06	0.004071	4.680986	COX7A1
10	0.321410	8.593706	4.827912	2.020835e-06	0.004093	4.591181	CHORDC1
11	0.723353	9.050568	4.791520	2.398513e-06	0.004453	4.438251	NaN
12	-0.223188	10.003463	-4.742596	3.014769e-06	0.005166	4.234261	ABR
13	-0.192992	10.850371	-4.666180	4.292557e-06	0.006830	3.919316	CLIP3
14	0.494327	10.691965	4.642281	4.789546e-06	0.007113	3.821742	CRYAB
15	0.184804	8.534250	4.620018	5.301965e-06	0.007216	3.731246	P4HA1
16	0.607572	9.316949	4.611698	5.506686e-06	0.007216	3.697522	HSPB1
17	-0.133025	8.632771	-4.580546	6.342736e-06	0.007850	3.571740	IFT122
18	0.168338	6.812448	4.492703	9.409093e-06	0.009932	3.221123	CXCL11
19	-0.108827	7.572047	-4.490770	9.490427e-06	0.009932	3.213477	ZNF512B

	logFC	AveExpr	t	P.Value	adj.P.Val	B	gene
0	-0.149755	7.644869	-5.475425	8.061401e-08	0.001122	7.571669	GRIK1
1	-0.361868	8.833856	-5.408818	1.139735e-07	0.001122	7.255710	NPTX2
2	-0.280916	9.512150	-5.354025	1.511559e-07	0.001122	6.998241	FAM131A
3	0.584476	11.992566	5.245582	2.625234e-07	0.001340	6.495245	GFAP
4	0.106553	8.353890	5.218613	3.007336e-07	0.001340	6.371514	SPR
5	0.135576	8.359324	5.132654	4.620038e-07	0.001715	5.980799	PHF20
6	0.092186	7.802221	4.887665	1.521836e-06	0.004601	4.897989	NaN
7	0.196235	7.817970	4.870378	1.652437e-06	0.004601	4.823324	NaN
8	-0.203078	9.799710	-4.732229	3.163685e-06	0.007831	4.234937	CYP46A1
9	-0.213419	9.798199	-4.673881	4.143263e-06	0.009230	3.990901	ARPC5L
10	-0.242092	9.568450	-4.640055	4.838552e-06	0.009799	3.850650	CCND2
11	-0.078719	7.341390	-4.612038	5.498168e-06	0.009920	3.735163	NaN
12	-0.288388	9.382766	-4.595238	5.934278e-06	0.009920	3.666209	MMD
13	-0.101358	7.797995	-4.584365	6.234037e-06	0.009920	3.621703	GGCX
14	-0.151356	8.754362	-4.543889	7.482999e-06	0.010425	3.456844	TMEM184B
15	0.515301	9.316949	4.543749	7.487709e-06	0.010425	3.456276	HSPB1
16	0.104271	8.169866	4.524819	8.151608e-06	0.010604	3.379618	NaN
17	-0.136184	9.508558	-4.513682	8.568205e-06	0.010604	3.334653	KCMF1
18	0.082030	7.506294	4.479507	9.977953e-06	0.011226	3.197289	EPHA3
19	-0.144877	9.214965	-4.477131	1.008382e-05	0.011226	3.187775	MGRN1

Figure 12: Contrasting gene expression results for brain region DLPFC for different age cutoffs. Genes in both lists: HSD11B1, PENK, RPS6KB1, XIST

	logFC	AveExpr	t	P.Value	adj.P.Val	B	gene
0	0.475154	8.143204	6.077507	3.032231e-09	0.000068	10.106294	PENK
1	0.163189	7.337408	5.353494	1.515686e-07	0.001688	6.692220	NaN
2	0.197900	7.141529	5.131241	4.652548e-07	0.003455	5.716845	NaN
3	-0.287624	8.042899	-5.036491	7.418131e-07	0.003475	5.311751	CACNA1G
4	0.199533	7.967097	5.021828	7.968511e-07	0.003475	5.249639	RPS6KB1
5	-0.252824	7.932824	-4.988733	9.359704e-07	0.003475	5.110023	NCRNA00185
6	0.400829	7.530007	4.841550	1.894647e-06	0.006030	4.498785	XIST
7	0.210548	8.067660	4.762622	2.746016e-06	0.007325	4.177553	RNF130
8	-0.232920	8.448161	-4.746594	2.959169e-06	0.007325	4.112883	ICA1
9	0.401348	8.335722	4.676149	4.100253e-06	0.008674	3.830914	FKBP4
10	0.211975	7.417980	4.666656	4.283180e-06	0.008674	3.793198	WDR33
11	0.155555	7.462614	4.627277	5.129358e-06	0.009522	3.637467	NBLA00301
12	0.166148	7.703911	4.588919	6.106767e-06	0.010465	3.486887	ADAM17
13	0.240666	8.293281	4.496096	9.267892e-06	0.014747	3.127088	TEK
14	-0.277883	8.118566	-4.479574	9.974991e-06	0.014808	3.063726	KIF1C
15	0.162898	7.659655	4.465124	1.063554e-05	0.014808	3.008482	CUZD1
16	-0.289780	7.878189	-4.309128	2.102318e-05	0.026170	2.422207	BIRC7
17	0.378365	10.036650	4.307778	2.114566e-05	0.026170	2.417216	CACYBP
18	-0.216096	7.741201	-4.268072	2.506980e-05	0.029394	2.271008	NNMT
19	0.299761	8.182930	4.251550	2.689982e-05	0.029962	2.210526	HSD11B1

	logFC	AveExpr	t	P.Value	adj.P.Val	B	gene
0	0.161599	7.967097	5.262606	2.408746e-07	0.004380	6.267445	RPS6KB1
1	0.333775	7.530007	5.165048	3.932501e-07	0.004380	5.842524	XIST
2	0.303216	8.143204	4.916823	1.323796e-06	0.009830	4.792084	PENK
3	-0.144774	7.814390	-4.841799	1.892416e-06	0.010539	4.483383	MED20
4	-0.187034	7.570615	-4.726800	3.244455e-06	0.014455	4.018220	SLC25A24
5	0.200301	8.798190	4.614752	5.430675e-06	0.020163	3.574409	GAK
6	0.130034	7.555921	4.526480	8.091151e-06	0.025750	3.231378	LOC100128612
7	0.203047	8.226708	4.449631	1.139032e-05	0.031718	2.937510	NaN
8	-0.178328	8.088473	-4.404032	1.392133e-05	0.034458	2.765256	GRK5
9	-0.185620	7.146078	-4.312827	2.069093e-05	0.046093	2.425465	CD24
10	-0.377523	8.151007	-4.231573	2.928299e-05	0.051709	2.128110	NR4A2
11	0.101679	7.337408	4.198823	3.363144e-05	0.051709	2.009695	NaN
12	0.231921	8.182930	4.171578	3.771163e-05	0.051709	1.911818	HSD11B1
13	-0.166803	7.710764	-4.162608	3.915541e-05	0.051709	1.879716	DMD
14	0.127993	7.688687	4.156193	4.021998e-05	0.051709	1.856798	NaN
15	-0.208668	8.872286	-4.148936	4.145753e-05	0.051709	1.830911	ASS1
16	-0.287060	7.760736	-4.145351	4.208225e-05	0.051709	1.818136	USP9Y
17	0.112993	7.093009	4.138337	4.333035e-05	0.051709	1.793175	C10orf12
18	0.180376	7.731305	4.134094	4.410244e-05	0.051709	1.778093	KIF25
19	0.167838	8.232360	4.101342	5.051908e-05	0.056271	1.662140	ZNF500

Figure 13: Contrasting gene expression results for brain region CB for different age cutoffs. Genes in both lists: HSPB1, SPR

8 Appendix

The GitHub repo containing all code for completing this project can be found here[1]

8.1 EDA

This section provides technical details of the expression-similarity analyses and their implications for data preprocessing.

[1] https://github.com/ChanChanMan18/STATS604_Project2/tree/main

8.1.1 Cosine Similarity Analysis

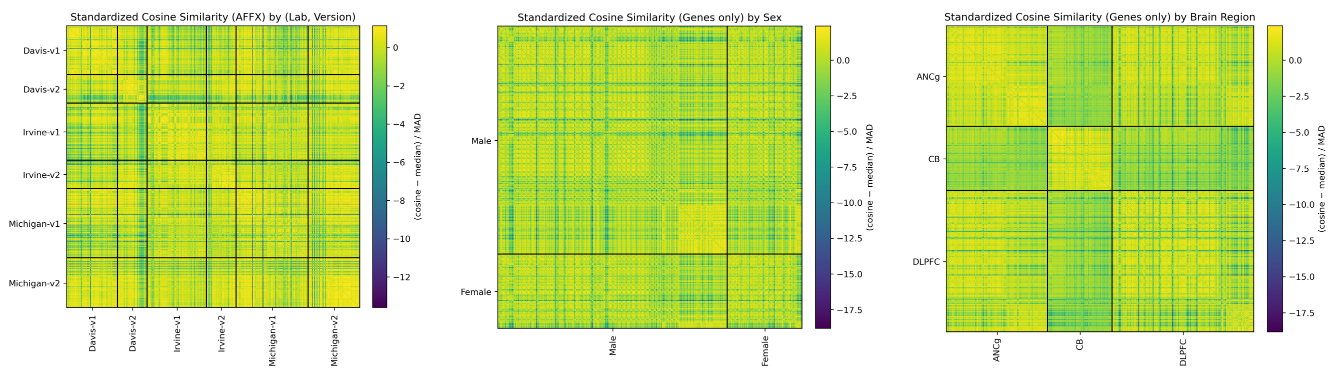
Pairwise similarity between sample expression vectors was quantified using the **cosine similarity** measure:

$$\text{cosine}(x_i, x_j) = \frac{x_i^\top x_j}{|x_i|_2 |x_j|_2}$$

where x_i and x_j denote the gene-expression vectors of two samples. This value ranges between 0 and 1, with 1 indicating perfect alignment (identical direction).

To assess technical versus biological structure:

- **Control-probe (AFFX)** expression matrices were used to estimate *technical variation*, free of biological signal.
- **All-gene** matrices (excluding AFFX) were used to capture total variability and potential biological clustering.



(a) AFFX by (lab,version)

(b) Non-AFFX by sex

(c) Non-AFFX by region

Figure 14: Standardized cosine-similarity heatmaps. Davis v2 shows distinct technical deviation; region clusters strongly; sex shows moderate structure.

8.1.2 Standardization of Cosine Similarities

Since cosine similarities are close to 1, raw values were standardized across all pairs:

$$z_{ij} = \frac{\text{cosine}(x_i, x_j) - \text{median}(\text{cosine})}{\text{MAD}(\text{cosine})}$$

where MAD is the mean absolute deviation. This transformation centers the distribution and highlights fine-scale deviations attributable to technical or biological structure.

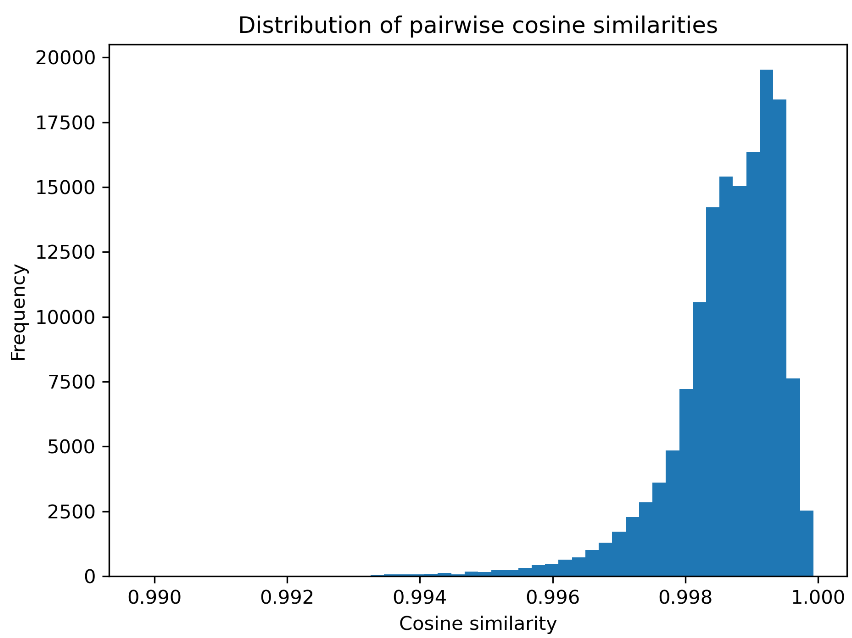


Figure 15: Left-skewed histogram of pairwise cosine similarities.

8.1.3 Distribution of Pairwise Similarities

Figure 15 shows the overall distribution of pairwise cosine similarities (all genes). The histogram is **strongly left-skewed**, with most values extremely close to 1. This indicates that samples are globally similar at a coarse scale, and subtle structure (lab, sex, region) lies in the small deviations from perfect alignment.

8.1.4 Interpretation

- Cosine similarities from control probes reveal that technical lab and array version effects dominate part of the observed structure.
- Region remains the most pronounced biological source of expression variability, while sex effects are moderate.

8.2 Description of Linear Modeling and Empirical Bayes

One tool applied for our analysis is the `limma` package in R, which provides an efficient implementation of the statistical framework developed in Smyth's paper "Linear Models and Empirical Bayes Methods for Assessing Differential Expression." The methodology introduced in the paper to identify DE genes is comprised of three main parts: first fitting a linear model to log gene expression, then applying a hierarchical Bayesian prior to stabilize gene-wise variance estimates, and finally performing multiple testing on the resulting "moderated" statistics. Such a procedure is necessary because with a large number of test statistics, we immediately run into a multiple testing problem. If we were to control for False Discovery Rate using Benjamini-Hochberg, for instance, we would arrive at very few significant genes. We detail how Smyth's procedure alleviates this problem below:

First, for each gene g, the observed log-expression value is modeled by a linear model of the form:

$$\mathbf{E}[y_g] = X\alpha_g \qquad\qquad \beta_g = C^\top \alpha_g$$

where y_g is the log expression value of gene g, X is the design matrix encoding the predictors (age group, sex, lab, brain region, array version), and α_g is the vector of regression coefficients specific to that gene. Thus, crucially, each gene has its own variance parameter σ_g^2 which will be estimated from the data. This is in contrast to the inverted design (e.g., using $\mathbb{1}[\texttt{age} \geq 70]$ as the response) where the genes would not have their own variance parameters. Also, note that the assumptions are presented this way instead of an explicit ϵ_g noise term to allow for greater generality in the linear model. The variable(s) of interest β_g are then given by $C^\top \alpha_g$ where C is some fixed vector. The estimators $\hat{\beta}_g$ and s_g (estimator of σ_g^2) are assumed to follow the distributions:

$$\hat{\beta}_g \mid \beta_g, \sigma_g^2 \sim \mathcal{N}(\beta_g, \sigma_g^2) \qquad\qquad s_g^2 \mid \sigma_g^2 \sim \frac{\sigma_g^2}{d_g}\chi^2_{d_g}$$

Next, to stabilize these variance estimates, Smyth imposes a hierarchical model. Specifically, the gene-specific variances are assumed to be drawn from a common prior distribution:

$$\frac{1}{\sigma_g^2} \sim \frac{1}{d_0 s_0^2}\chi^2_{d_0} \qquad\qquad \mathbf{P}(\beta_g \neq 0) = p \qquad\qquad \beta_g \mid \sigma_g^2, \beta_g \neq 0 \sim \mathcal{N}(0, v_0\sigma_g^2)$$

To estimate the parameters d_0, s_0^2, v_0, and p, Smyth uses Empirical Bayes. Essentially, we conduct Bayesian inference, but instead of using a fully specified prior, we estimate the parameters of the prior using some method (such as MLE). Smyth showed that, in this case, "the empirical Bayes approach is equivalent to shrinkage of the estimated sample variances towards a pooled estimate":

$$\tilde{s_g}^2 = \frac{d_0 s_0^2 + d_g s_g^2}{d_0 + d_g} \qquad\qquad \tilde{t}_g = \frac{\hat{\beta}_g}{\tilde{s}_g\sqrt{v_g}}$$

This hierarchical structure allows the variances across genes to "borrow strength" from one another, shrinking unstable estimates toward a common value. The moderated t-statistics derived from these posterior variances yield more reliable inference while still accommodating gene-to-gene variability.

8.3 Age-Predictive Linear Modelling Workflow and OLS Results

This section provides the detailed modeling workflow, parameter choices, and outcomes for the **age-predictive linear modeling** analysis described in the executive summary.

8.3.1 Modeling Objective

The goal of the modeling framework was to identify genes and covariates that are **predictive of chronological age**, under the hypothesis that such variables would also exhibit **age-dependent differential expression**. By formulating age as a response variable, this predictive modeling approach provides a structured and quantitative route for prioritizing biologically relevant features.

8.3.2 Workflow

1. **Train–Test Split** The dataset was divided into training and test subsets to avoid overfitting and to enable out-of-sample validation of model-selected covariates.

2. **Correlation-Based Screening** Within the training data, 234 genes were retained whose absolute correlation with age exceeded **0.4**. This moderate threshold was chosen to capture genes showing meaningful but not overly strict linear associations, preserving flexibility for complex biological patterns.

3. **LASSO Regression with Interaction Terms** A LASSO regression model was trained using five-fold cross-validation with a grid of 40 tuning parameters: $\alpha \in 10^{-4}, 10^{-4+1/40} \ldots, 10^{1}$. The predictors included both the screened genes and their **interaction terms** with categorical covariates from the array metadata (e.g., lab and array version indicators). After model selection, **118 covariates** were retained — comprising **37 individual genes** and the remainder being interaction terms.

4. **OLS Regression on the Test Set** The selected covariates were then used to fit a standard **ordinary least squares (OLS)** regression model on the held-out test set, treating chronological age as the dependent variable. The resulting model achieved a high $R^2 = 0.990$ and adjusted $R^2 = 0.921$, with an F-statistic of 14.37 (p = 4.06e5). These metrics indicate that the selected features collectively explain a large portion of age-related variability while maintaining generalization capability. The model summary is presented in Figure 16.

5. **Significance Testing and Correction** Using the fitted model coefficients, **Benjamini–Hochberg (BH)** correction (at $\alpha = 0.1$) was applied to identify significantly age-associated covariates. The significant covariates—both individual genes and interaction terms—were subsequently tested for **differential expression** using two-sample t-tests within each brain region. For consistency, BH correction ($\alpha = 0.1$) was again applied to the p-values from these regional tests.

6. **Interpretation** The model framework provides a cohesive link between predictive modelling and inferential DE analysis, reducing false positives by integrating variable selection, regularization, and multiple testing control. It allows for discovery of both globally predictive and context-specific (interaction-driven) age-associated expression patterns.

8.3.3 OLS Regression Results

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.990
Model:                            OLS   Adj. R-squared:                  0.921
Method:                 Least Squares   F-statistic:                     14.37
Date:                Wed, 08 Oct 2025   Prob (F-statistic):           4.06e-05
Time:                        14:23:21   Log-Likelihood:                -138.71
No. Observations:                  78   AIC:                             413.4
Df Residuals:                      10   BIC:                             573.7
Df Model:                          67
Covariance Type:            nonrobust
```

Figure 16: OLS Regression Results for the age-predictive model on the test set

The OLS fit confirms strong linear dependence between predicted and observed age values, validating that the LASSO-selected genes and their technical interactions capture substantial variance associated with aging.

The subsequent BH-based screening of coefficients yielded a focused set of candidate genes for downstream differential expression testing, from which the dorsolateral prefrontal cortex (DLPFC) emerged as the most strongly age-associated region.

8.3.4 Age predictive Linear Modeling with Age Cutoff=65

Differential Expression by Region (BH significant)

Gene	Probe	Region	Context	n<65	n≥65	t	p	q
ERBB2IP	217941_s_at	DLPFC	arrayv=1 | subset: arrayversion=1	88	32	-4.477	0.0	0.001
PLSCR4	218901_at	DLPFC	sex=M	127	51	-4.631	0.0	0.001
PTPRO	211600_at	DLPFC		127	51	-3.981	0.0	0.004
SLC7A11	209921_at	DLPFC	sex=M	127	51	-3.776	0.0	0.004
nan	208451_s_at	DLPFC		127	51	-3.81	0.0	0.004
FAM107A	209074_s_at	DLPFC	arrayv=1 | subset: arrayversion=1	88	32	-3.757	0.0	0.004
FAM107A	209074_s_at	DLPFC	arrayv=1	127	51	-3.809	0.0	0.004
ERBB2IP	217941_s_at	DLPFC	arrayv=1	127	51	-3.603	0.0	0.005
PLSCR4	218901_at	DLPFC	sex=M | subset: sex=M	101	31	-3.685	0.001	0.005
PPAP2B	212226_s_at	DLPFC	sex=M	127	51	-3.632	0.0	0.005
QKI	212636_at	DLPFC	sex=M	127	51	-3.522	0.001	0.005
RHOQ	212119_at	DLPFC		127	51	-3.538	0.001	0.005
SLC7A11	209921_at	DLPFC	sex=M | subset: sex=M	101	31	-3.605	0.001	0.005
PPAP2B	212226_s_at	DLPFC	sex=M | subset: sex=M	101	31	-3.55	0.001	0.005
IGF1R	203627_at	DLPFC		127	51	-3.356	0.001	0.007
NFE2L2	201146_at	DLPFC	sex=M	127	51	-3.315	0.001	0.007
FTL	212788_x_at	DLPFC	region.dlpfc	127	51	-3.274	0.001	0.008
IFI16	206332_s_at	DLPFC	sex=M	127	51	-3.278	0.001	0.008
CHI3L1	209395_at	DLPFC		127	51	-3.219	0.002	0.008
HSPB8	221667_s_at	DLPFC		127	51	-3.211	0.002	0.008

Figure 17: Significant genes and covariates identified from the age-predictive linear model with cutoff=65

8.4 Description of Classification Modeling

We use 5-fold CV to determine the penalty parameter and the classification threshold. Notice that the two classes are imbalanced. As a result, besides the overall accuracy, we also look at the ROC curve in Figure 18 and choose the threshold accordingly. Table 19 shows the confusion matrix of the classification result. We can see that the misclassifications are spread across two groups as desired.

The top 20 genes with cutoffs 65 and 70 are shown in Figures 21 and 20.

To identify the region to follow up on, we note that conditional model (or simply include region:gene interaction term) is feasible in this approach, but whether the main effect or the interaction is more preferable remains unclear and could be misleading. For example, if the interaction term appears significant but the main effect does not, ranking according to either one does not seem sufficient.

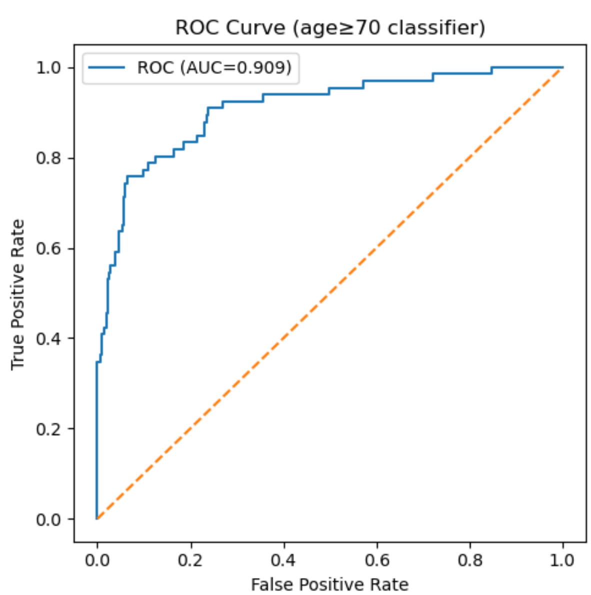


Figure 18: ROC curve for the ridge logistic classifier on age 70.

	Pred 0 (< 70)	Pred 1 (≥ 70)
True 0 (< 70)	311	9
True 1 (≥ 70)	29	37

Figure 19: Confusion matrix for the ridge logistic classifier on age 70.

Rank	Gene ID	Symbol	|Beta|
1	203815_at	GSTT1	0.2768
2	202917_s_at	S100A8	0.2249
3	209619_at	CD74	0.1856
4	211719_x_at	FN1	0.1724
5	211990_at	HLA-DPA1	0.1707
6	202581_at	N/A	0.1692
7	205966_at	TAF13	0.1632
8	204712_at	WIF1	0.1597
9	204570_at	COX7A1	0.1575
10	210949_s_at	N/A	0.1555
11	211959_at	IGFBP5	0.1519
12	208894_at	HLA-DRA	0.1512
13	212464_s_at	FN1	0.1504
14	206190_at	GPR17	0.1491
15	203540_at	GFAP	0.1453
16	219750_at	TMEM144	0.1452
17	216598_s_at	CCL2	0.1433
18	215230_x_at	N/A	0.1396
19	210495_x_at	FN1	0.1392
20	210982_s_at	HLA-DRA	0.1390

Figure 20: Top 20 genes for the ridge logistic classifier on age 70.

Rank	Gene ID	Symbol	|Beta|
1	205048_s_at	PSPH	0.3093
2	204712_at	WIF1	0.2319
3	213979_s_at	N/A	0.1958
4	213812_s_at	CAMKK2	0.1876
5	210910_s_at	POMZP3	0.1688
6	208719_s_at	DDX17	0.1654
7	219750_at	TMEM144	0.1567
8	204148_s_at	N/A	0.1542
9	218469_at	GREM1	0.1528
10	207526_s_at	IL1RL1	0.1521
11	203540_at	GFAP	0.1476
12	202203_s_at	AMFR	0.1465
13	201123_s_at	EIF5A	0.1463
14	206190_at	GPR17	0.1446
15	214012_at	ERAP1	0.1397
16	215470_at	N/A	0.1390
17	208691_at	TFRC	0.1377
18	221540_x_at	N/A	0.1352
19	217753_s_at	RPS26	0.1325
20	202917_s_at	S100A8	0.1319

Figure 21: Top 20 genes for the ridge logistic classifier on age 65.